

IT 462:
Big Data Systems
Course Project
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Diabetes Health Indicators

Phase 3

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Section#:	2766		
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1. Overview

In this phase, Apache Spark RDD operations were applied to the preprocessed CDC Diabetes Health Indicators dataset to explore patterns and derive meaningful insights. The dataset was transformed into a structured RDD format, enabling efficient distributed computation across multiple analytical tasks. A combination of transformations and actions was used to extract trends related to diabetes risk, lifestyle factors, and population health.

2. RDD Transformations and Actions

2.1 Dataset Size Verification

Action: count

The total number of records in the dataset was computed using the count() action to verify successful data loading and preprocessing.

Sample Output:

```
2.1 Dataset Size Verification
Total number of records: 229500
```

Interpretation:

The dataset contains 229,500 records after preprocessing, ensuring sufficient scale for meaningful analysis in a distributed Spark environment.

2.2 Data Validation Sample

Action: first

```
2.2 Data Validation Sample
First record:
0.0, 1.0, 1.0, 25.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0, 0.0, 1.0, 0.0, 2.0, 0.0, 0.0, 1.0, 1.0, 11.0, 2.0, 3.0, 2.0, 1.0
```

A sample record was retrieved using first() to verify the correctness of parsing and data structure before performing aggregations.

Sample Output:

```
2.2 Data Validation Sample
First record:
0.0, 1.0, 1.0, 25.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0, 0.0, 1.0, 0.0, 2.0, 0.0, 0.0, 1.0, 1.0, 11.0, 2.0, 3.0, 2.0, 1.0
```

Interpretation:

This step confirms that all 22 attributes are correctly parsed and typed. Validating the data structure at this stage is essential, as incorrect parsing would propagate errors into all subsequent transformations.

2.3 Diabetes Rate by Age Group

Transformations: map, reduceByKey, mapValues, sortBy

Actions: take, foreach

Individuals were grouped by age category using map and reduceByKey. mapValues computed the diabetes rate per group. Results were sorted by age group and displayed using take and foreach.

Sample Output:

2.3 Diabetes Distribution by Age Group			
Age Group	Diabetes Cases	Total	Diabetes Rate (%)
Age 1	177	5502	3.22%
Age 2	332	7057	4.70%
Age 3	697	9988	6.98%
Age 4	1390	12220	11.37%
Age 5	2256	14026	16.08%
Age 6	3787	17275	21.92%
Age 7	6556	23114	28.36%
Age 8	9012	27260	33.06%
Age 9	12060	29708	40.60%
Age 10	13639	29138	46.81%
Age 11	10775	22027	48.92%
Age 12	7204	15381	46.84%
Age 13	6850	16804	40.76%

Interpretation:

The results show a clear upward trend in diabetes prevalence with age. Younger groups (Age 1–3, representing 18–34) exhibit low rates of approximately 7–9%, while the oldest groups (Age 12–13, representing 75+) exceed 38–40%. This demonstrates a strong positive relationship between age and diabetes risk, highlighting the importance of targeted screening for older populations.

2.4 Average BMI by Diabetes Status

Transformations: map, reduceByKey, mapValues

Action: collect

Records were keyed by diabetes status using map. reduceByKey accumulated total BMI and count, and mapValues computed the mean. Results were retrieved with collect.

Sample Output:

2.4 Average BMI by Diabetes Status	
Diabetes Status	Average BMI
No Diabetes	27.96
Prediabetes	30.68
Diabetes	31.90

Interpretation:

Individuals diagnosed with diabetes and prediabetes have a notably higher average BMI (31.90) and (30.68) respectively compared to non-diabetic individuals (27.96). This finding reinforces obesity as a major risk factor and supports preventive strategies focused on weight management.

2.5 Identification of High-Risk Individuals

Transformation: filter

Action: count

A filter transformation was applied to isolate individuals with all four combined risk factors: high blood pressure (HighBP=1), smoking history (Smoker=1), physical inactivity (PhysActivity=0), and fair-to-poor general health (GenHlth ≥ 3).

Sample Output:

2.5 Identification of High-Risk Individuals	
High-risk individuals count: 14313	
Percentage of total population: 6.24%	

Interpretation:

Over 14,000 individuals (6.24% of the dataset) simultaneously exhibit high blood pressure, a smoking history, physical inactivity, and poor self-reported health. This clustering of risk factors highlights the importance of integrated intervention strategies rather than addressing single risk factors in isolation.

2.6 BMI Category Distribution

Transformations: map, reduceByKey, sortBy

Actions: take, foreach

BMI values were mapped to standard clinical categories (Underweight, Normal, Overweight, Obese) using a map transformation. reduceByKey computed category frequencies, and sortBy ranked them by descending count.

Sample Output:

2.6 BMI Category Distribution		
BMI Category	Count	Percentage
Obese	84661	36.89%
Overweight	82859	36.10%
Normal	58927	25.68%
Underweight	3053	1.33%

Interpretation:

The majority of the population falls in the overweight (36.10%) and obese (36.89%) categories, meaning over 77% of individuals are above a healthy BMI. This distribution directly aligns with the observed association between elevated BMI and diabetes prevalence.

2.7 General Health Distribution

Transformations: map, reduceByKey, sortByKey

Actions: collect, foreach

Self-reported general health scores (1–5) were aggregated using map and reduceByKey, then sorted in ascending order by key using sortByKey and retrieved with collect.

Sample Output:

2.7 General Health Distribution

GenHlth	Label	Count	Percentage
1	Excellent	34857	15.19%
2	Very Good	77433	33.74%
3	Good	73634	32.08%
4	Fair	31519	13.73%
5	Poor	12057	5.25%

Interpretation:

The largest share of respondents rated their health as Very Good (33.74%) or Good (32.08%). However, around 19% of individuals rated their health as Fair or Poor, indicating a substantial portion of the population with meaningful underlying health concerns, which is consistent with the broader context of chronic disease prevalence.

2.8 Top Age Groups by Diabetes Prevalence

Transformation: sortBy

Action: take

The age group RDD computed in Section 2.3 was re-sorted in descending order by diabetes rate using sortBy, and the top 5 groups were extracted with take(5).

Sample Output:

2.8 Top Age Groups by Diabetes Prevalence

Rank	Age Group	Diabetes Rate	Cases
1	Age 11	48.92%	10775/22027
2	Age 12	46.84%	7204/15381
3	Age 10	46.81%	13639/29138
4	Age 13	40.76%	6850/16804
5	Age 9	40.60%	12060/29708

Interpretation:

The five age groups with the highest diabetes rates are all the oldest categories in the dataset (Age 9–13, corresponding to individuals aged 60 and above). Age 13 (80+) has the highest rate at 40.03%, further confirming age as one of the strongest predictors of diabetes risk in this dataset.

3. Summary

3.1 Summary of Insights

The RDD-based analysis reveals several important patterns:

- Diabetes prevalence increases significantly with age, rising from ~7% in the 18–24 age group to over 40% in the 80+ group.
- Higher BMI is strongly associated with diabetes: diabetic individuals have an average BMI of 31.90 versus 27.96 for non-diabetic individuals.
- Multiple lifestyle risk factors (high BP, smoking, physical inactivity, poor health) frequently coexist, affecting over 14,000 individuals.
- Over 70% of the population falls into the overweight or obese BMI categories, placing them at elevated chronic disease risk.
- Nearly 20% of respondents self-report Fair or Poor general health, reflecting widespread underlying health concerns.

These findings directly support the project’s objective of identifying at-risk individuals using non-clinical data. By leveraging Spark’s distributed processing capabilities, the analysis demonstrates how large-scale health datasets can be efficiently explored to uncover actionable insights for early detection and prevention.

3.2 Summary of RDD Actions

RDD Action	Section Used	Purpose
count	2.1, 2.5	Verify dataset size; count high-risk individuals
first	2.2	Retrieve and validate a single parsed record
take	2.3, 2.6, 2.8	Extract top N results for display
collect	2.4, 2.7	Retrieve all key-value results to driver
foreach	2.3, 2.6, 2.7	Iterate and print results to console

3.3 Summary of RDD Transformations

RDD Transformation	Section Used	Purpose
map	2.3, 2.4, 2.6, 2.7	Parse records; create key-value pairs
filter	2.5	Isolate high-risk individuals by criteria
reduceByKey	2.3, 2.4, 2.6, 2.7	Aggregate counts and sums by key
mapValues	2.3, 2.4	Compute rates and averages per group
sortBy	2.6, 2.8	Rank results by value (descending)
sortByKey	2.7	Sort GenHlth distribution by score
distinct	Preprocessing	Remove duplicate records if present

4. Conclusion

This phase successfully demonstrates the effective use of Spark RDD transformations and actions for large-scale data analysis. The operations performed were both computationally meaningful and directly aligned with the problem statement. The insights derived provide strong evidence that lifestyle, demographic, and self-reported health data can be used to identify individuals at risk of diabetes, supporting data-driven decision-making in public health.

Each operation in this phase included a clearly labelled sample output block and a separate interpretation paragraph, ensuring reproducibility and transparency of results.

5. References

- [1] Centers for Disease Control and Prevention, “About Adult BMI,” 2023. [Online]. Available: <https://www.cdc.gov/bmi/adult-calculator/index.html>. [Accessed: Feb. 27, 2026].